

20260617EB_MN (Glo, ATP, NCT, IF)

Project: Yi-Shuan Tseng Holt Lab

Author: Yi Shuan Tseng

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2026年6月17日 星期三

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2026年6月19日 星期五

T2 -> T3

2026年6月21日 星期日

T3 -> T4 (~14 mL for over-weekend)

2026年6月24日 星期三

T4 -> T5

2026年6月26日 星期五

T5 -> T6(~14mL for over-weekend)

2026年6月29日 星期一

T6 -> T6

Coat Laminin

2026年6月30日 星期二

T6 -> T7

Dissociation

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 3mL 0.25% trypsin for each (prepare 8mL 0.25% trypsin with 220U DNaseI)
3. Incubate the tubes in 37C.
4. pipette*10 at 5min, 10min.
5. Stop reaction at ~12 (A), 13 (B) min by adding 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
6. add 8mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.
8. centrifuge at 200xg, 4min

cell info ^

	A	B	C	D
1		cell	passage	well#
2	A	WT11	P19+2	3
3	B	WT11	P19+2	4

Table1 ^

	A	B
1		volume (mL)
2	trypsin	4
3	HI-FBS/CTWM	8

Table3 ✓

	A	B	C	D	E	F	G	H	I	J
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	A	3.31	100	5	16.55	I (A)	96	100	30	906
3	B	4.88	100	5	24.4	I (B)	96	100	30	615
4						II (A) 92, IF	24	300	8	725
5						II (B) 92, IF	24	300	8	492
6						III (A) 110, 111	24	300	4	363
7						III (B) 110, 111	24	300	4	246

Plate I ^

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

Plate II						
	1	2	3	4	5	6
A	110 (1uL/well)				no virus	
B	111 (1.6uL/well)					
C	110 (1uL/well)				no virus	
D	111 (1.6uL/well)					

Plate III						
	1	2	3	4	5	6
A	92 (1uL/well)				no virus	
B						
C	92 (1uL/well)				no virus	
D						

2026年7月1日 星期三

Change T7 -> T7
Cells survived a lot and looked very healthy

2026年7月3日 星期五

T7 -> BP T8 (new media)

2026年7月5日 星期日

1/2 T8 -> 1/2 T8

2026年7月7日 星期二

1/2 T8 -> 1/2 T8
Thaw CellTiter-Glo reagent in 4C

2026年7月8日 星期三

Cells are healthy!!!

CellTiter Glo

- 1. Prepare 1400uL media for each and add 200uL in two 96 well plate (one for wash one for treatment). 1 condition has 6 wells. (6*200+200 = 1400)
- 2. remove original media with multichannel pipette

3. wash with the media
4. add media and incubate for 1h

Table4

	A	B	C	D
1	condition #			
2	1	2.5mM gluc + 1mM pyruvate	6	0mM MMS
3	2	0.15mM gluc + 1mM pyruvate	7	0.5mM MMS
4	3	0mM glucose + 1mM pyruvate +FCCP+2DG	8	0.75mM MMS
5	4	0.15mM gluc + 5mM pyruvate	9	1mM MMS
6	5	0.15mM gluc + 10mM pyruvate	10	2mM MMS

Plate I_condition

	1	2	3	4	5	6	7	8	9	10	11	12
A	1mM			100uM			1uM			100nM		
B	C	1	2	3	4	5	1	2	3	4	5	E
C												
D												
E	D	6	7	8	9	10	6	7	8	9	10	
F												F
G												
H	G			0mM gluc+1mM pyruvate			2.5mM gluc+1mM pyruvate			blank		

prepare ATP calibration							
	A	B	C	D	E	F	G
1	A	B	C	D	E	F	G
2	100mM (stock)	1mM	1uM	300nM	100nM	30nM	10nM
3	dilution factor from the previous one	100	1000		10		10
4					from C		fromE
5		4uL A in 399uL media	1uL B in 999uL	180uL C in 420uL media	50uL C in 450uL media	150uL D in 350uL media	50uL D in 450uL media

prepare media with different conc. of glucose					
	A	B	C	D	E
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)
2	1100	2.5	440	9000	20.5
3	2.5	0.15	16.6666666667	4500	270.0
4				13500	-> 14mL + 140uL pyruvate

prepare media with different conc. of glucose with pyruvate							
	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	2DG	300	6	50	1400	28.0	1372.0
3	FCC P	1	0.001	1000	1400	1.4	1398.6

drug calculation													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution (mM)	working (mM)	dilution factor from 1st dil.	total volume (mL)	volume of 1st dil.	volume of media
2	MMS	6							0		1400	0	1000
3		7	11800	100	100	1	99	118	0.5	236	1400	5.9	1394.1
4		8	11800	100	100	1	99	118	0.75	157.3333333333	1400	8.9	1391.1
5		9	11800	100	100	1	99	118	1	118	1400	11.9	1388.1
6		10	11800	100	100	1	99	118	2	59	1400	23.7	1376.3
7										total	7000	50.4	

Table2			
	A	B	C
1		14:00	prepare media
2	-105	14:40	prepare media into a 96-well plate
3	-75	14:50-15:00	add media to cells
4	-23	15:50	prepare ATP standard and add samples
5	-17	16:00	add celltiter glo reagent
6	-12	16:05	put the plate into cytation 5 and start program to shake
7	-10	16:07	wait
8	0	16:17	measure

Imaging NCT batch A (MMS and olaparib)

1. Prepare 9mL 2.5mM gluc NBA + T8 supp + 1mM pyruvate
2. Take 8mL and add 1.3uL SirDNA (1:6000)
3. Right before doing DMSO experiment
 - a. Dilute DMSO (10mM stock) 1:10 (1uL into 9uL media)
 - b. Take 4mL media and add 0.8uL DMSO
4. Dilute MMS 1uL in 99uL media right before the experiment
5. Prepare different concentration of MMS
6. Mix well
7. Right before doing olaparib experiment
 - a. Dilute olaparib (10mM stock) 1:10 to 1mM (1uL into 9uL media)
 - b. Take 4mL media and add 0.8uL 1mM olaparib

O.C. , 60X, 600x900, 16-bit, New Spinning Disk Ti2					^
	A	B	C	D	
1	channel	bit	power (%)	exposure time	
2	640	16	10	100ms	
3	561	16	25 (110)/30 (111)	300ms	
4	488	16	50	1s	

NCT conditions							^
	1	2	3	4	5	6	
A	MMS 0mM +DMSO	MMS 1mM + DMSO	MMS 0mM +olaparib	MMS 1mM +olaparib			
B	MMS 0mM +DMSO	MMS 1mM + DMSO	MMS 0mM +olaparib	MMS 1mM +olaparib			
C	MMS 0mM +DMSO	MMS 1mM + DMSO	MMS 0mM +olaparib	MMS 1mM +olaparib			
D	MMS 0mM +DMSO	MMS 1mM + DMSO	MMS 0mM +olaparib	MMS 1mM +olaparib			

drug calculation1														/
	A	B	C	D	E	F	G	H	I	J	K	L	M	
1			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution (mM)	working (mM)	dilution factor from 1st dil.	total volume (mL)	volume of 1st dil.	volume of media	
2	MM S	1							0		1000	0	1000	
3		2	11800	100	100	1	99	118	1	118	2000	17.1	1982.9	
4			11800	100	100	1	99	118	0.5	236	2000	8.5	1991.5	
5			11800	100	100	1	99	118	2	59	2000	34.5	1965.5	

time table for NCT treatment_MMS_Plate II-BatchA										
	A	B	C	D	E	F	G	H	I	J
1	NCT (plate I)									
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	
3										
4	A	A1	MMS 0mM	NT	2.5mM gluc	1mM	1	16:50	17:48	
5	B	A2	MMS 1mM	NT	2.5mM gluc	1mM	3	18:10	19:08	
6	A'	A3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	5	19:35	20:32	
7	B'	A4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	7	20:55	21:55	
8										
9	A	B1	MMS 0mM	NT	2.5mM gluc	1mM	2	17:30	18:28	
10	B	B2	MMS 1mM	NT	2.5mM gluc	1mM	4	18:49	19:47	
11	A'	B3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	6	20:06	21:08	
12	B'	B4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	8	21:30	22:30	

fileID_treatment_NCT_20260708_MN_110,111						
	A	B	C	D	E	F
1	image	reporter	treatment	drug	time	pyruvate
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	DMSO	1h	yes
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	DMSO	1h	yes
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	1mM MMS	DMSO	1h	yes
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	1mM MMS	DMSO	1h	yes
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0mM MMS	200nM olaparib	1h	yes
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	0mM MMS	200nM olaparib	1h	yes
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	1mM MMS	200nM olaparib	1h	yes
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	1mM MMS	200nM olaparib	1h	yes
10	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	DMSO	1h	yes
11	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	DMSO	1h	yes
12	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	1mM MMS	DMSO	1h	yes
13	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	1mM MMS	DMSO	1h	yes
14	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0mM MMS	200nM olaparib	1h	yes
15	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	0mM MMS	200nM olaparib	1h	yes
16	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	1mM MMS	200nM olaparib	1h	yes
17	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	1mM MMS	200nM olaparib	1h	yes

2026年7月9日 星期四

Imaging NCT batch B (MMS and olaparib)

time table for NCT treatment_MMS_Plate II-BatchB										
	A	B	C	D	E	F	G	H	I	J
1	NCT (plate I)									
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	
3										
4	1	A1	MMS 0mM	NT	2.5mM gluc	1mM	1	10:50	12:14	delayed 25min
5	2	A2	MMS 1mM	NT	2.5mM gluc	1mM	3	12:37	13:37	
6	3	A3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	5	13:57	14:59	
7	4	A4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	7	15:20	16:29	delayed 9min
8										
9	1	B1	MMS 0mM	NT	2.5mM gluc	1mM	2	10:18	10:16	
10	2	B2	MMS 1mM	NT	2.5mM gluc	1mM	4	12:00	12:58	
11	3	B3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	6	13:20	14:20	
12	4	B4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	8	14:40	15:40	

fileID_treatment_NCT_20260709_MN_110,111						
	A	B	C	D	E	F
1	image	reporter	treatment	drug	time	pyruvate
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	DMSO	1h	yes
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	DMSO	1h	yes
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	1mM MMS	DMSO	1h	yes
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	1mM MMS	DMSO	1h	yes
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0mM MMS	200nM olaparib	1h	yes
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	0mM MMS	200nM olaparib	1h	yes
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	1mM MMS	200nM olaparib	1h	yes
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	1mM MMS	200nM olaparib	1h	yes
10	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	DMSO	1h	yes
11	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	DMSO	1h	yes
12	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	1mM MMS	DMSO	1h	yes
13	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	1mM MMS	DMSO	1h	yes
14	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0mM MMS	200nM olaparib	1h	yes
15	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	0mM MMS	200nM olaparib	1h	yes
16	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	1mM MMS	200nM olaparib	1h	yes
17	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	1mM MMS	200nM olaparib	1h	yes

Perceval

I noticed that NCT did not change with 1mM MMS for NCT. I suspected that it might because the MMS was aliquoted on 7/7/2026 (on Tuesday)

I decided to make a comparison between the aliquot in use and a new aliquot from the stock.

Well1						
	1	2	3	4	5	6
A	1	2	3	4		
B	5	6	7	8		
C	1	2	3	4		
D	5	6	7	8		

time table for ATP treatment_MMS_Plate III_BatchA,B1											
	A	B	C	D	E	F	G	H	I	J	K
1	NCT (plate I)										
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	note	note2
3											
4	1	A1	MMS 0mM	DMSO	2.5mM gluc	1mM	1	16:00	17:00		
5	2	A2	MMS 1mM	DMSO	2.5mM gluc	1mM	2	16:20	17:20		
6	3	A3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	3	16:35	17:35		
7	4	A4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	4	16:55	17:55		
8											
9	1	C1	MMS 0mM	DMSO	2.5mM gluc	1mM	1	17:03	18:11	8 min late	
10	2	C2	MMS 1mM	DMSO	2.5mM gluc	1mM	2	17:20	17:28	8 min late	I used a fresh aliquot for batchB
11	3	C3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	3	17:40	18:39		
12	4	C4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	4	18:00	19:00		

fileid_treat_20260709_MN_92						
	A	B	C	D	E	F
1	image	treatment	drug	treat_time	pyruvate	replicate
2	vLH92_MN_60x-1xzoom_0000	0mM MMS	DMSO	1h	yes	A
3	vLH92_MN_60x-1xzoom_0001	1mM MMS	DMSO	1h	yes	A
4	vLH92_MN_60x-1xzoom_0002	0mM MMS	200nM olaparib	1h	yes	A
5	vLH92_MN_60x-1xzoom_0003	1mM MMS	200nM olaparib	1h	yes	A
6	vLH92_MN_60x-1xzoom_0004	0mM MMS	DMSO	1h	yes	B
7	vLH92_MN_60x-1xzoom_0005	1mM MMS	DMSO	1h	yes	B
8	vLH92_MN_60x-1xzoom_0006	0mM MMS	200nM olaparib	1h	yes	B
9	vLH92_MN_60x-1xzoom_0007	1mM MMS	200nM olaparib	1h	yes	B

2026年7月10日 星期五

1/2 T8 -> 1/2 T8

Perceval

Note: The media used to treat Batch B was contaminated. I saw lots of rod bacteria in bright field. I prepared new media for batch A and it looks clear.

time table for ATP treatment_Plate III_BatchA,B2												
	A	B	C	D	E	F	G	H	I	J	K	L
1	NCT (plate I)											
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	note		
3												
4	1	B1	MMS 0mM	DMSO	2.5mM gluc	1mM	5	11:45	12:45	new media was prepared		
5	2	B2	MMS 1mM	DMSO	2.5mM gluc	1mM	6	12:00	13:01			
6	3	B3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	7	12:23	13:23			
7	4	B4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	8	12:50	13:50			
8												
9	1	D 1	MMS 0mM	NT	2.5mM gluc	1mM	1	10:00	11:10	treating media was contaminated		
10	2	D 2	MMS 0.5mM	NT	2.5mM gluc	1mM	2	10:30	11:32			
11	3	D 3	MMS 1mM	NT	2.5mM gluc	1mM	3	10:50	11:55			
12	4	D 4	MMS 2mM	NT	2.5mM gluc	1mM	4	11:13	12:15			

fileid_treat_20260709_MN_1						
	A	B	C	D	E	F
1	image	treatment	drug	treat_time	pyruvate	replicate
2	vLH92_MN_60x-1xzoom_0000	0mM MMS	NT	1h	yes	B
3	vLH92_MN_60x-1xzoom_0001	0.5mM MMS	NT	1h	yes	B
4	vLH92_MN_60x-1xzoom_0002	1mM MMS	NT	1h	yes	B
5	vLH92_MN_60x-1xzoom_0003	2mM MMS	NT	1h	yes	B
6	vLH92_MN_60x-1xzoom_0004	0mM MMS	DMSO	1h	yes	A
7	vLH92_MN_60x-1xzoom_0005	1mM MMS	DMSO	1h	yes	A
8	vLH92_MN_60x-1xzoom_0006	0mM MMS	200nM olaparib	1h	yes	A
9	vLH92_MN_60x-1xzoom_0007	1mM MMS	200nM olaparib	1h	yes	A

IF:

Treat cells at 13:50-56, 14:05-14:10

Fix cells at 14:50

Plate II IF conditions

	1	2	3	4	5	6
A					2.5mM gluc (set A)	0.15mM gluc (set A)
B					0.15mM gluc (set B)	2.5mM gluc (set A)
C					2.5mM gluc (set B)	0.15mM gluc (set A)
D					0.15mM gluc (set A)	2.5mM gluc (set A)

Plate III IF conditions

	1	2	3	4	5	6
A					0mM MMS	0.5mM MMS
B					1mM MMS	2mM MMS
C					0mM MMS	0.5mM MMS
D					1mM MMS	2mM MMS

antibody

	A	B	C	D	E	F	G	H	I	J	K	L
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	C-31	RANGAP1	Rb	500	720	1.44	488	B	1000	2880	2.88
3	A	146	Ran	m	500	720	1.44	594	O	1000	2880	2.88
4	B	from Mazzone lab	SOD1	Rb	1000	720	0.72	488				
5	B	147	Importin-b1	m	1000	720	0.72	594				
6	C	63	H2A.X (Ser139) clone JBW301	Mouse mono	2000	1440	0.72	488				
7	C	102	TIA1	Rb	500	1440	2.88	594				

IF imaging

Use Nikon Ti2 in Innolab

SetA OC 60x,600x900, 16bit					^
	A	B	C	D	
1		laser power (%)	exposure time (ms)		
2	DIC			BF	
3	405	10	50	DAPI	
4	488	38	200	RANGAP1	
5	561	30	300	Ran	
6	640	30	500	Halodye (TDP43-Halo)	
7	z-stack	1um/step	15um	17 steps in total	

SetB OC 60x,600x900, 16bit						^
	A	B	C	D	E	
1		laser power (%)	exposure time (ms)			
2	DIC			BF		
3	405	10	50	DAPI		
4	488	50	500	SOD1		
5	561	30	500	Impb1		
6	640	30	500	Halodye (TDP43-Halo)		
7	z-stack	1um/step	15um	17 steps in total	*the slices below -5 are usually no cells	

fileID_treatment_IF									
	A	B	C	D	E	F	G	H	I
1	image	treatment	treat_time	media	replicate	488	561	640	
2	A5_488-RanGAP1_561-Ran_640-Halo_405-DAPI	2.5mM gluc	1h	full	A	RanGAP1	Ran	TDP43Halo	
3	A6_488-RanGAP1_561-Ran_640-Halo_405-DAPI	0.15mM gluc	1h	full	A	RanGAP1	Ran	TDP43Halo	
4	D5_488-RanGAP1_561-Ran_640-Halo_405-DAPI	0.15mM gluc	1h	full	B	RanGAP1	Ran	TDP43Halo	I accidentally turned the incubator light on for the first run. I re-took the images again.
5	D6_488-RanGAP1_561-Ran_640-Halo_405-DAPI	2.5mM gluc	1h	full	B	RanGAP1	Ran	TDP43Halo	
6	B5_488-SOD1_561-Impb1_640-Halo_405-DAPI	0.15mM gluc	1h	full	A	SOD1	Impb1	TDP43Halo	
7	B6_488-SOD1_561-Impb1_640-Halo_405-DAPI	2.5mM gluc	1h	full	A	SOD1	Impb1	TDP43Halo	
8	C5_488-SOD1_561-Impb1_640-Halo_405-DAPI	2.5mM gluc	1h	full	B	SOD1	Impb1	TDP43Halo	
9	C6_488-SOD1_561-Impb1_640-Halo_405-DAPI	0.15mM gluc	1h	full	B	SOD1	Impb1	TDP43Halo	